PATIENT DEMOGRAPHICS

- **Patient name**
- **DOB**
- **Sex**
- **Hospital / NHS number**

RADIOGRAPH DETAILS

- **Date and Time**

- **Orientation:**
  - Supine AP / Lateral Decubitus:
    - Assume Supine AP unless stated
  - Lateral Decubitus used to detect Pneumoperitoneum when patient unable to tolerate erect CXR

- **Adequacy:**
  - Should see whole abdomen, including:
    - Lung bases
    - Pelvis

RADIOGRAPH INTERPRETATION

A Air | **B** Bowel | **C** Calcification | **E** Extra | **S** Soft Tissue

Air (Intraperitoneal Free Air)

Intraperitoneal Free Air:
- Air around the bowel
- Should NOT be present:
  - Unless recent abdominal surgery
- Best assessed for along with erect CXR
  - Or Lateral Decubitus if patient unable to tolerate sitting upright for >15 minutes

Rigler's sign (Double wall sign) :
- Air on both sides of the bowel wall

Football sign:
- Massive pneumoperitoneum where abdomen appears 'blown-up like a football'

Erect CXR:
- Must be upright for > 15 minutes
- Air under diaphragm - Crescent shaped
  - **Note:** normal to have gastric air bubble under left diaphragm
  - **Note:** Bowel can reside quite high and be directly under diaphragm on right side. However air will be within bowel and not crescent shaped

Lateral Decubitus:
- Must lie on side for > 15 minutes
- Air under anterior wall of abdomen

Bowel Gas

Stomach:
- Only visible if has air in it
- Under left diaphragm

Liver / Common Bile Duct:
- No air should be visible
  - Unless previous endoscopic sphincterotomy

Small Intestine:
- Central in film:
  - Can have loops under right diaphragm (see above)
- **Valvular conniventes** (plicae circularis):
  - Mucosal folds that **cross the whole of the lumen**/diameter of bowel

Large Intestine:
- Travels around central bowel loops
- **Haustra** (taenia coli):
  - Transverse and longitudinal muscles which form outpouchings from bowel wall which **do not cross entire lumen**
- Mottled appearance - combination of faeces and air
- Caecum is largest and should be confined to RIF

Rectum:
- Should have air / faeces within it

Bowel Obstruction

Bowel diameter 3 / 6 / 9 rule:
- Small bowel 3cm / Large bowel 6cm / Caecum 9cm
  - Bowel diameters greater than these suggest obstruction distal to that point

Comma Sign:
- Caecal volvulus
- Dilated caecum more centrally located

Coffee Bean Sign:
- Sigmoid volvulus

Empty Rectum:
- Air in rectum suggests AGAINST bowel obstruction

Bowel Inflammation

Toxic Megacolon:
- Huge dilated large bowel due to Inflammation and NOT obstruction

Thumb Printing:
- Thickened inflammed haustra start to look like thumbs sticking into the bowel

Lead piping:
- Loss of haustra usually in chronic Ulcerative Colitis

Air in Liver

Pneumobilia:
- Air in CBD (Central liver):
  - Previous endoscopic sphincterotomy
  - Cholecysto-enteric fistula in cases of gallstone ileus

Portal Venous Gas:
- Air in Portal Vein (Periphery of the liver):
  - Generally represents life-threatening intra-abdominal pathology such as bowel ischaemia or necrosis

Calcifications

Calcifications /Stones:
- Normally none are seen
- Common to have Phleboliths and some Vascular calcification in increasing age
  - Vascular calcification should be linear tram-track

Renal calculi:
- Ideally assessed with CT KUB
- However always check for renal calculi on supine AXR:
  - Ureters pass down the lateral margins of the transverse processes of the lumbar vertebrae

Gallstones:
- Only 15% are Radio-opaque:
  - Thus look for **Pneumobilia** as sign of Gall stone ileus

Renal Calculi:
- Majority are radio-opaque
- Staghorn Calculi within renal pelvis
- Ureteric calculi often found:
  - Pelvicoureteric junction
  - Vesicoureteric junction
  - Within bladder (having passed through ureter)

Uroliths (Bladder stones):
- Majority are radio-opaque
- Tend to be larger than renal Calculi

Appendicolith:
- Calcified stone in appendix highly suggestive of appendicitis

Vascular:
- Sign of atherosclerosis
- Fusiform shape rather than the normal linear tram-track calcification suggestive of aneurysmal artery

Extras

Lung Bases:
- Check lung bases for evidence of opacity

Artifacts:
- Medical lines:
  - NG tubes, NJ tube, Urinary catheter
- Surgery:
  - Urinary / colonic stents
  - Clips
  - Drains
- Other:
  - Contraception devices
  - Jewellery (belly-button piercing) - confused as calculi

Basal Pneumonia:
- Can cause acute epigastric pain

NG Tube / NJ Tube / Urinary Catheter Position:
- Check correct position

Drain / Stent Position:
- Many types with specific locations
- Discuss with senior if concerned

Soft Tissue + Bones

Soft Tissues:
- Should be reasonably uniform in density
- Liver
- Kidney:
  - T12-L3 level
  - Right lower than Left
- Psoas muscles:
  - Lateral to lumbar transverse processes
  - Lateral border visible due to adjacent fat

Bones:
- Should have clear outlines
- Equal on both sides
- Rib / Spinal / Pelvic / Sacrum / Femur

Soft-tissue / Organ Increased Opacity:
- Potential mass

Psoas Sign:
- Loss of lateral edge of one / both psoas muscle(s)
  - Non-specific sign of potential retroperitoneal pathology

Ascites:
- General 'hazy' appearance of AXR

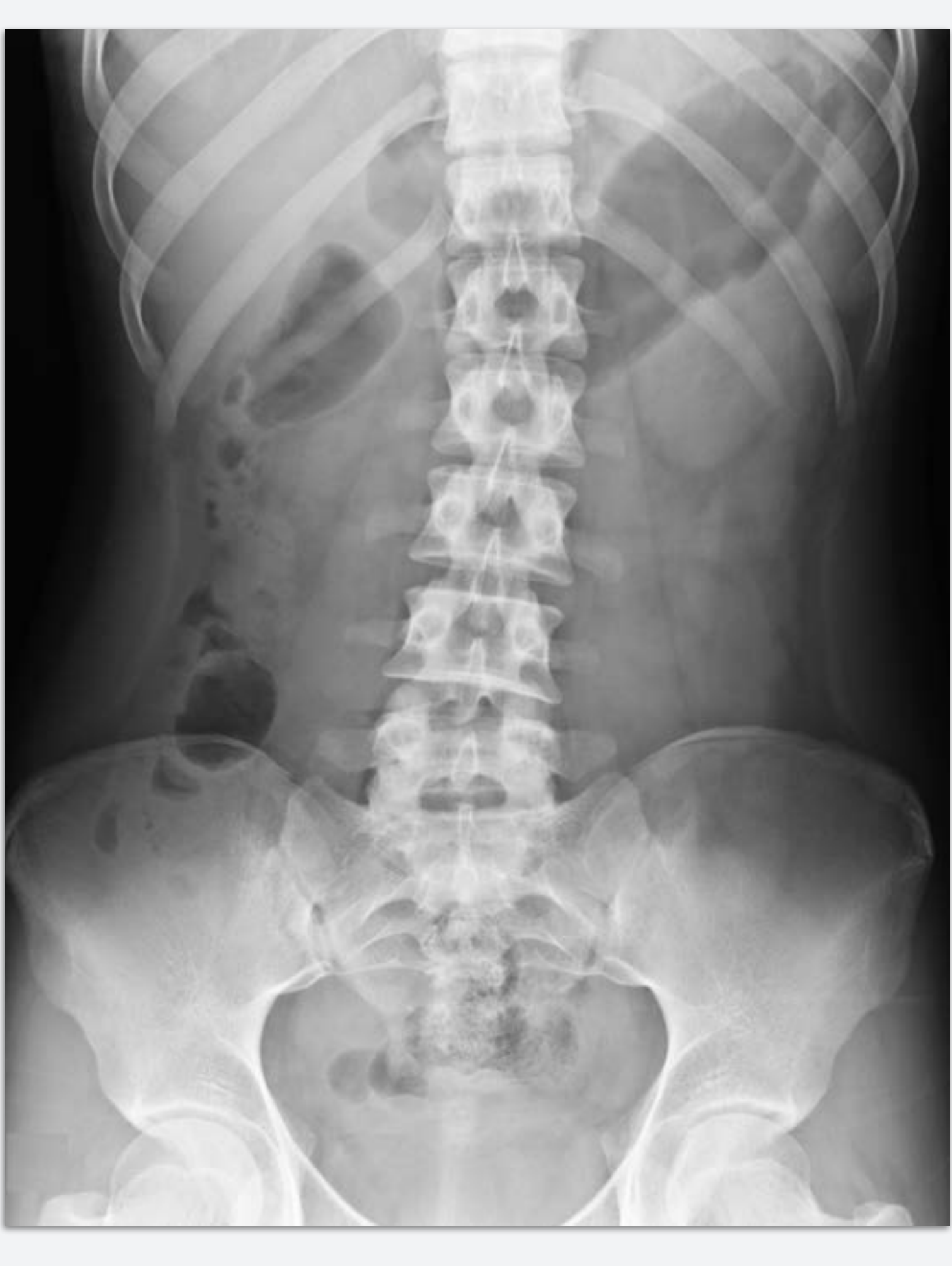
Bone irregularities :
- Fractures
- Primary / Metastatic lesion: Lytic (darker) Sclerotic (whiter)

Final Points

- **Comparison:**
  - If available compare to previous X-ray
- **Focus:**
  - Examine whole X-ray, don't focus on obvious abnormality
- **Patient:**
  - Always interpret X-ray with clinical picture in mind

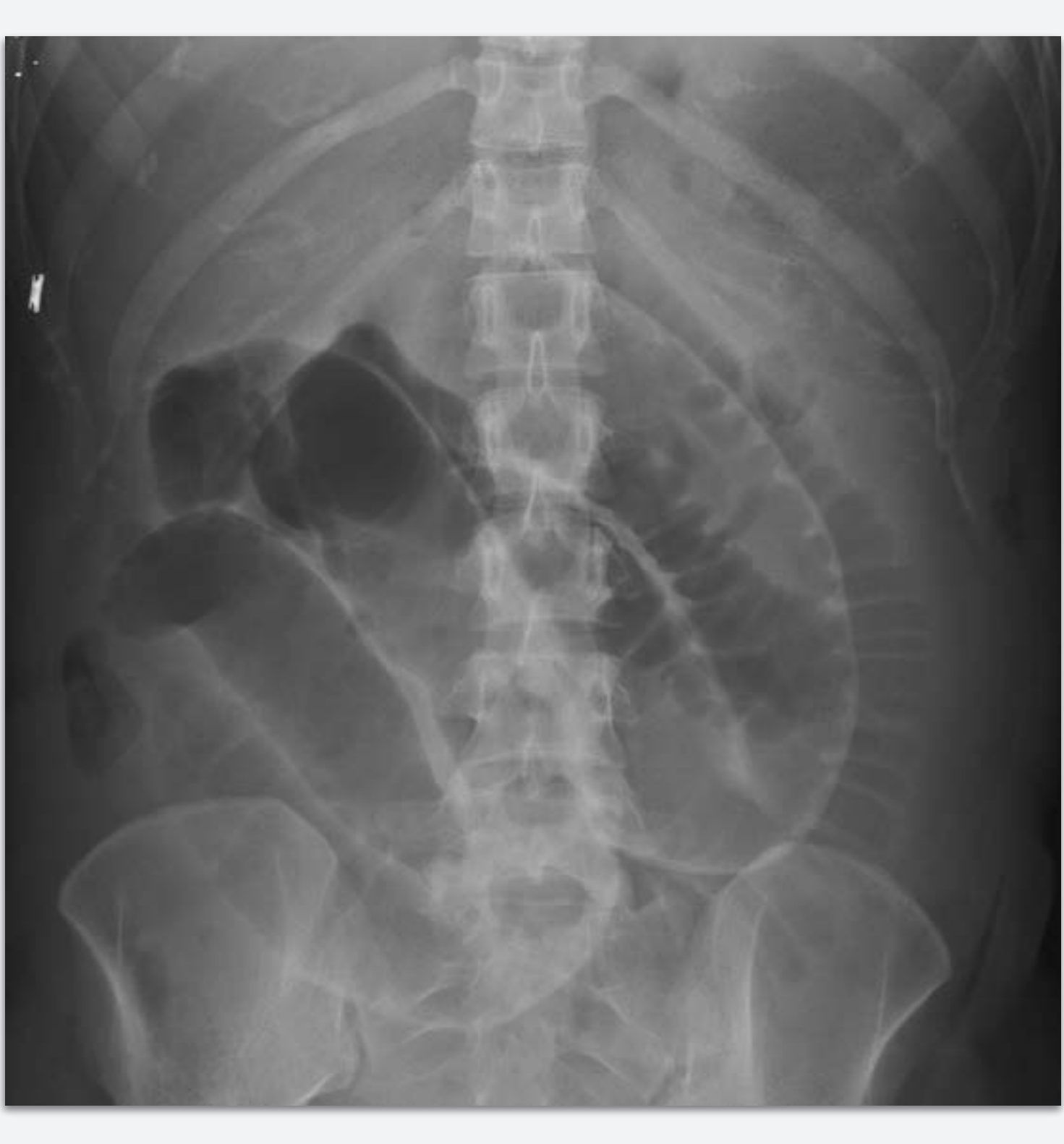
NORMAL AXR

Normal AXR

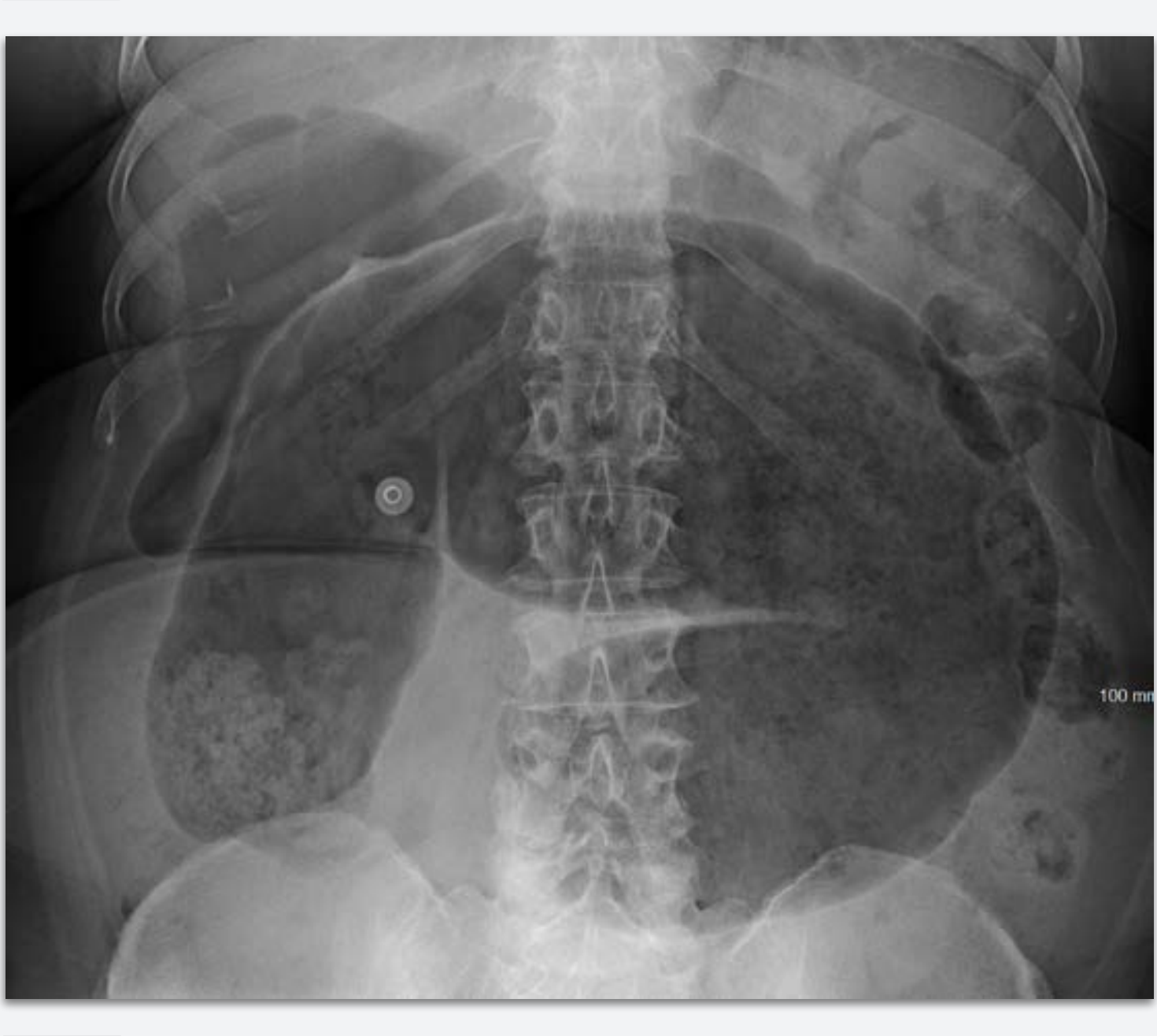


COMMON FINDINGS

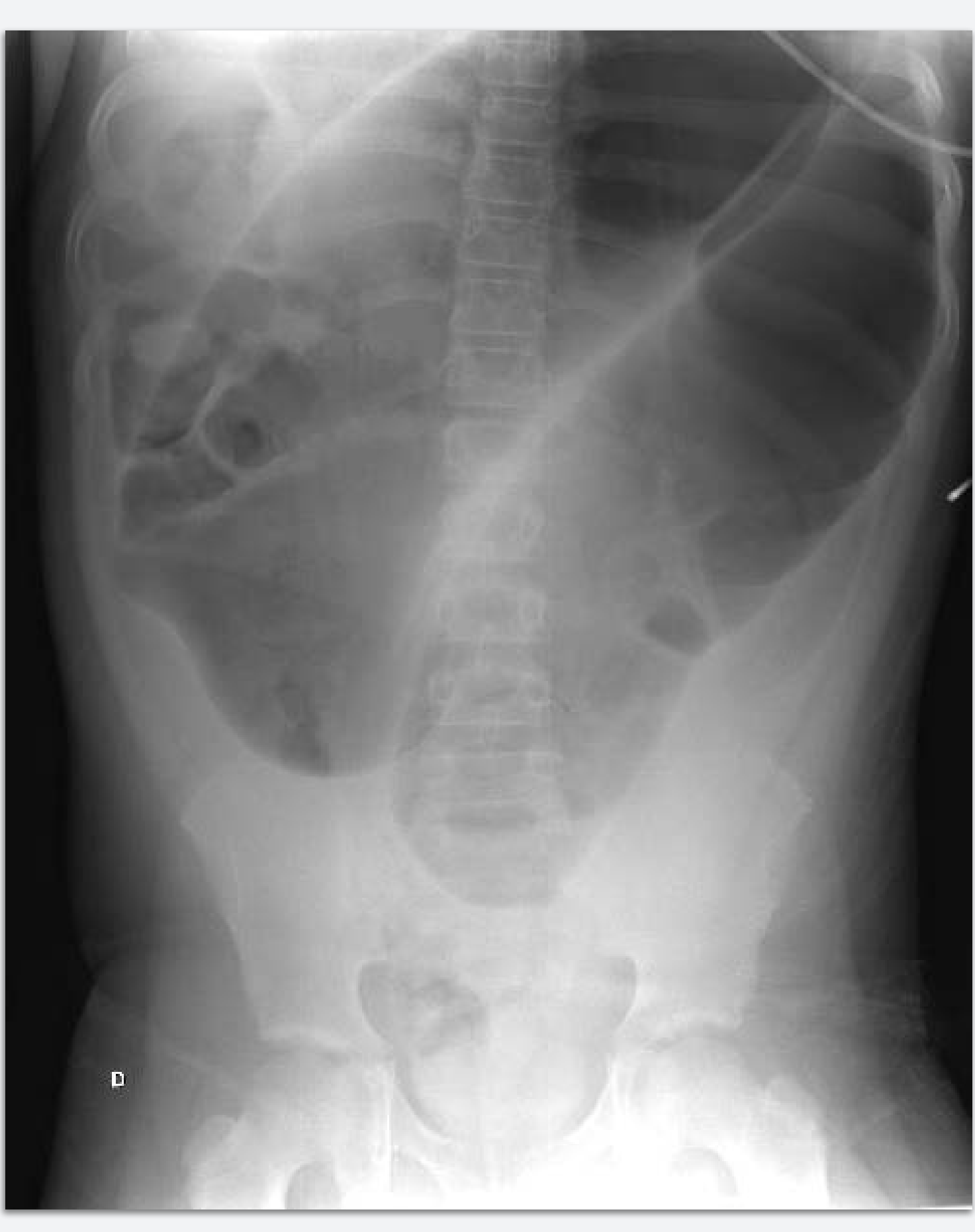
Small Bowel Obstruction:
- Dilated small bowel loops



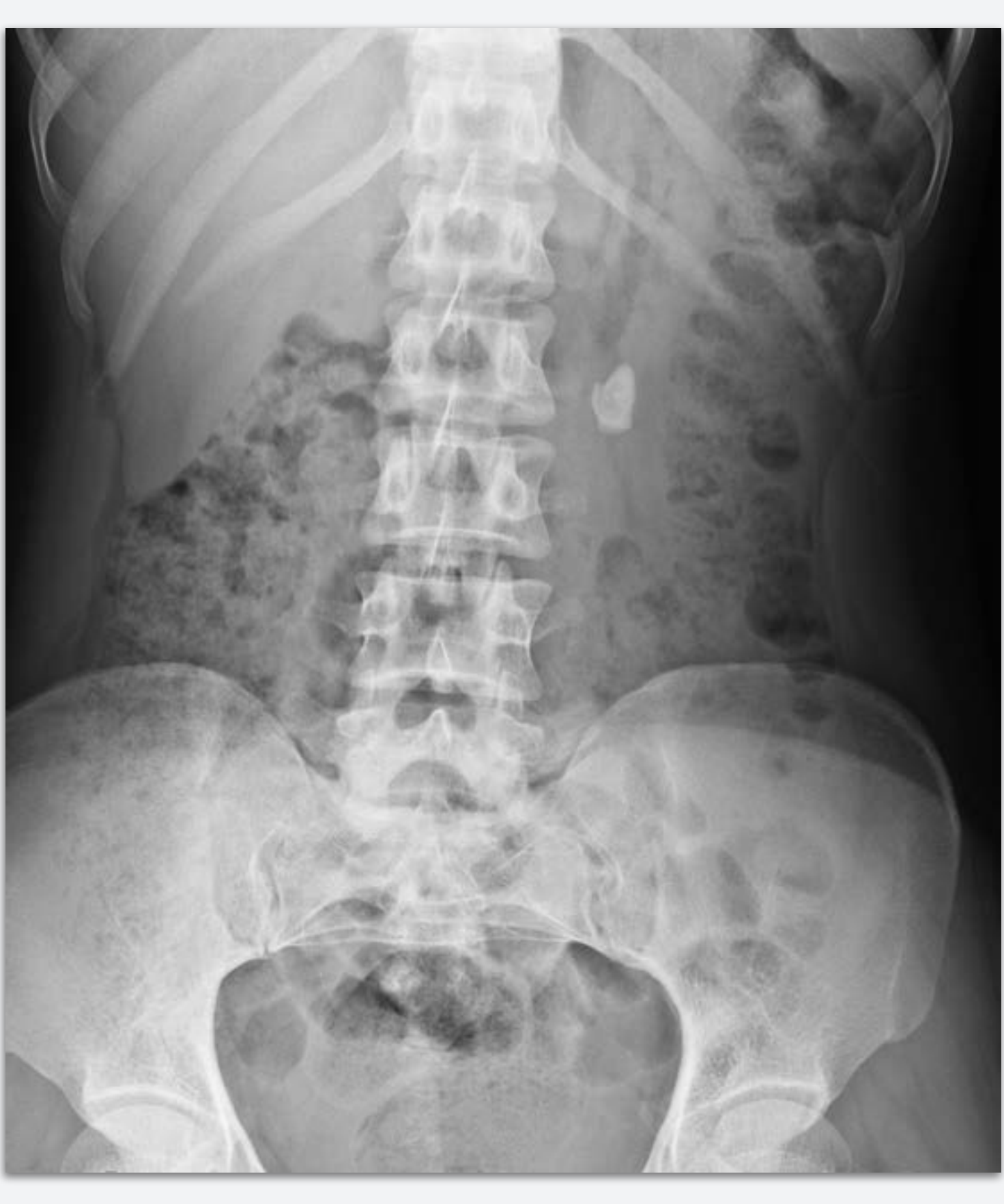
Caecal Volvulus / Large Bowel Obstruction:
- Comma sign / Dilated large bowel



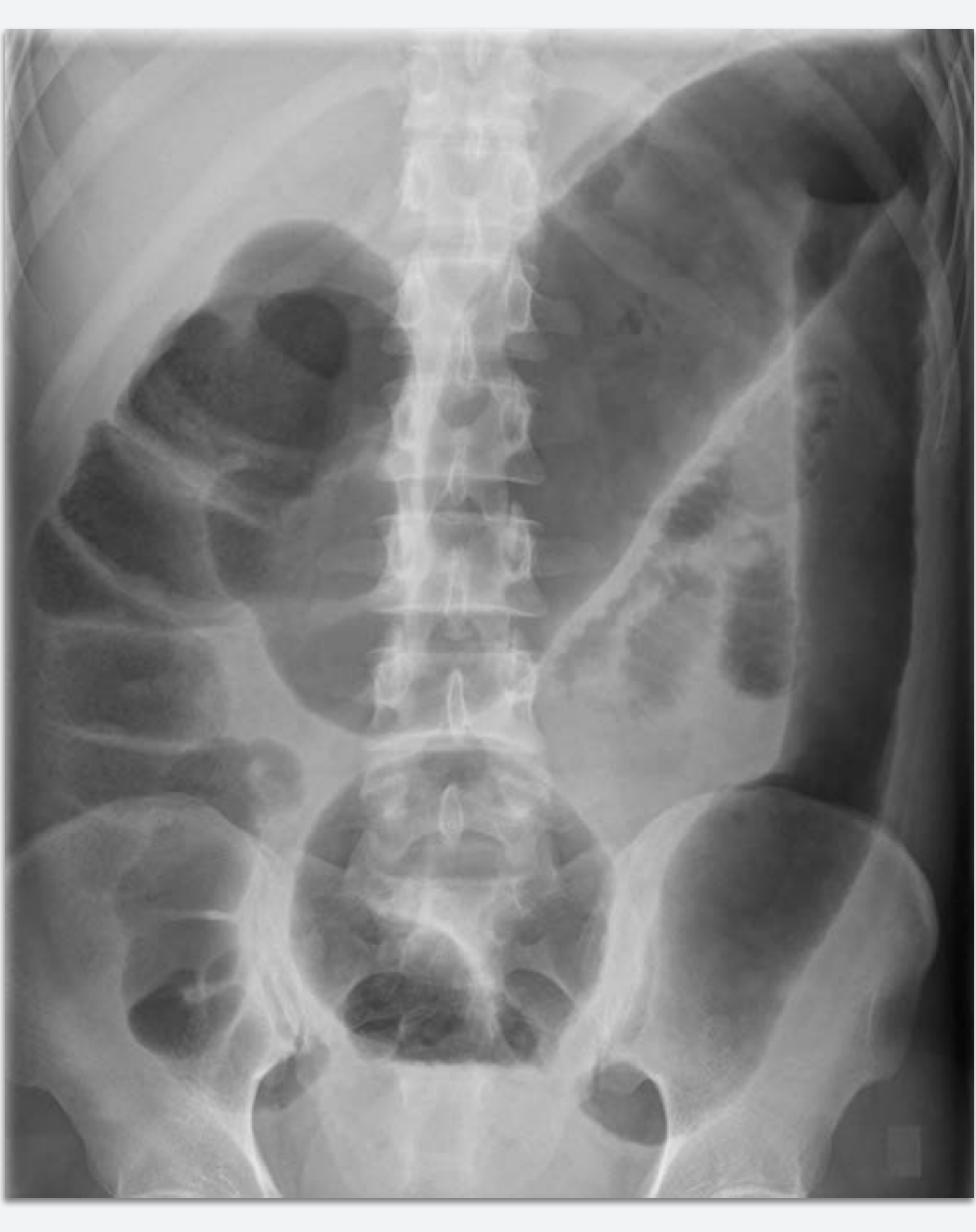
Sigmoid Volvulus / Large Bowel Obstruction:
- Coffee bean sign / Dilated large bowel



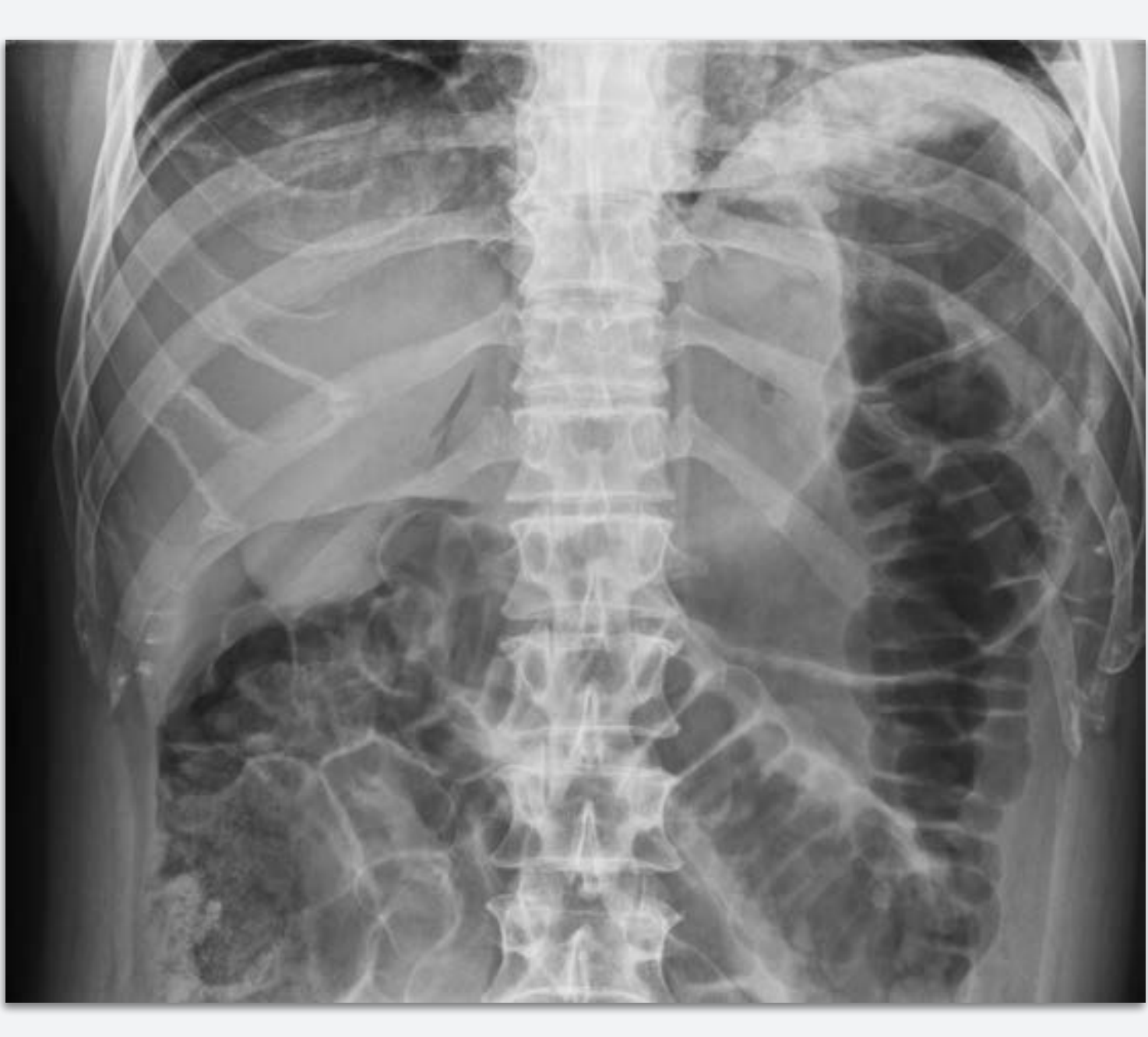
Renal Stones:
- Left pelvic renal stone

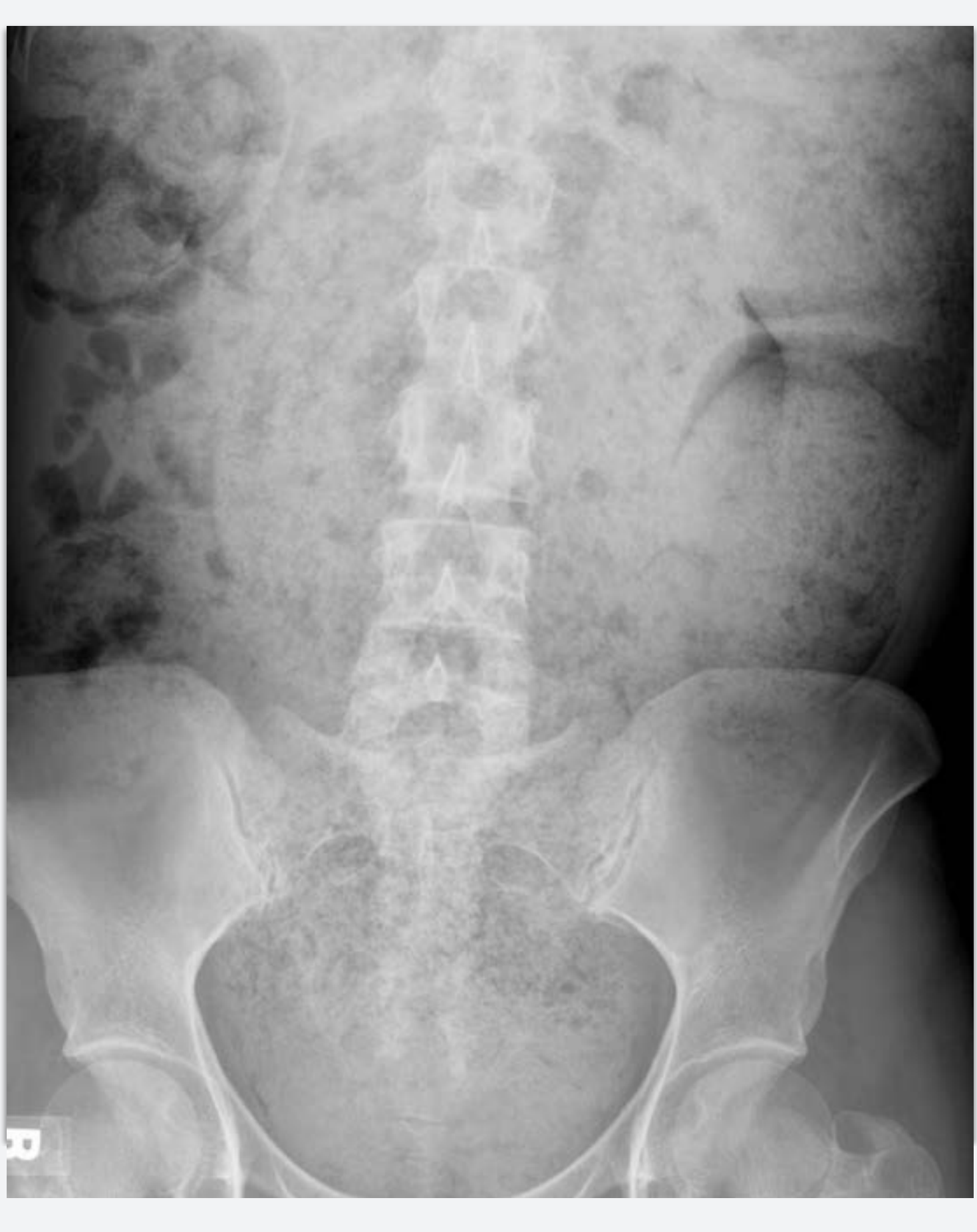


Megacolon / Inflammatory:
- Dilated large bowel + Lead pipe appearance



Pneumoperitoneum:
- Riglers sign - see both sides of bowel wall



Constipation / Faecal Loading:
- Mottled gas pattern throughout